

Bayesian Network Analysis for Loss of Controlled Flight (LCF)

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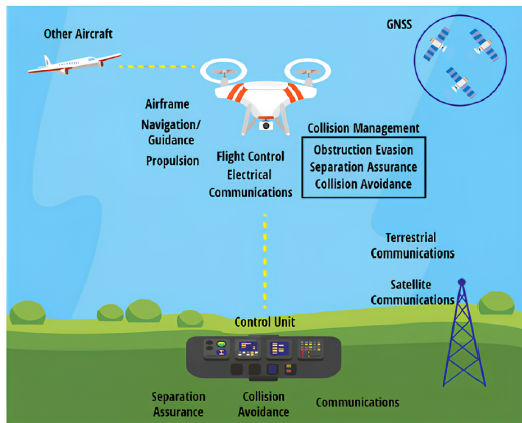
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1. System Overview

- Bayesian Network (BN) developed to estimate the probability of **Loss of Controlled Flight (LCF)** in a UAS.
- Structure derived from a Fault Tree in the reference article.
- Includes software faults, environmental hazards, communication issues, and human errors.



2. Model Structure

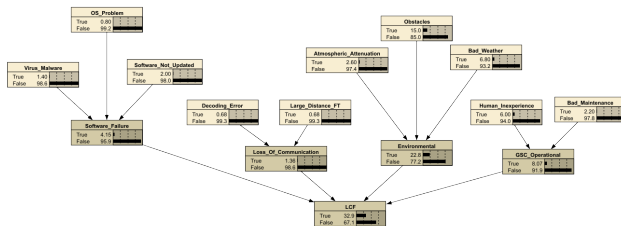
- **Basic events:** OSProblem, VirusMalware, Obstacles, BadWeather, HumanInexperience, etc.
- **Intermediate nodes:**
 - SoftwareFailure
 - LossOfCommunication
 - Environmental
 - GSCOperational
- **Top event:** LCF
- All CPTs modelled using **Noisy-OR**.
- Prior probabilities extracted from literature.

3. A Priori Analysis

- Computed unconditional probability:

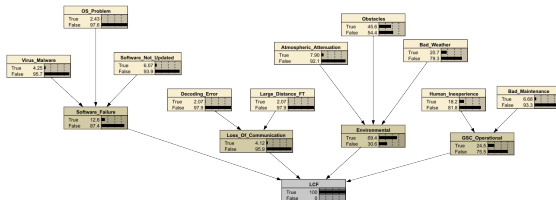
$$P(\text{LCF} = \text{True}) \approx 0.329$$

- Environmental hazards are the dominant contributors.
- Provides baseline for posterior, sensitivity, and reliability analyses.



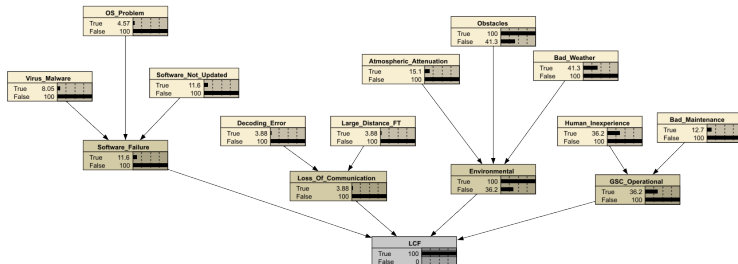
4. A Posteriori Analysis

- Conditioning on **LCF = True**.
- Posterior probabilities of intermediate nodes:
 - Environmental: 0.694
 - GSCOperational: 0.245
 - SoftwareFailure: 0.126
 - LossOfCommunication: 0.041
- Environmental hazards and operator errors emerge as main contributors.



5. Most Probable Explanation (MPE)

- MPE = most probable joint configuration of states given **LCF = True**.
- Scenario dominated by environmental hazards, with operator factors as supporting contributors.



6. Reliability and Conclusions

- Failure rate:

$$\lambda \approx 0.399 \text{ failures/mission}$$

- MTBF:

$$\text{MTBF} \approx 2.5 \text{ missions}$$

- Key insights:

- Environmental hazards are the primary drivers of LCF.
- Human factors also significantly contribute.
- Improvements needed: sensing, obstacle detection, operator training.